Reciprocal Traces and the Half-Filled Plunge Law

Kirtus Justus (Theoretical Physicist)
Port Coquitlam, British Columbia, Canada

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Abstract

For the half-filled finite sine kernel at even size, the first reciprocal trace of the deflated differential spectrum is an exact positive Laplace transform. A weighted Legendre resolvent and a Mellin contour identity determine its normalization. Taylor estimates on a fixed analytic kernel then prove its large-size expansion to every fixed order, including an explicit remainder after the term of order N^{-9}. This result yields the signed plunge law for every nonzero Jacobi eigenmode whose eigenvalue tends to zero. An elementary trial vector proves that the central pair has this property. No uniform bulk quantum-period expansion is assumed.

1 The finite problem and its exact differential trace

Let $N \geq 2$ be even. Define real symmetric matrices, with indices $0 \leq j, k < N$, by

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad H_{j-1,j} = H_{j,j-1} = \frac{j(N-j)}{2}, \quad H_{jj} = 0, \qquad J_N = \frac{2H}{N}.$$

These are the usual half-filled concentration matrix and its commuting Jacobi matrix [2]. They commute, $0 < B < I$, and their common eigenpairs are denoted

$$Hv = \beta v, \qquad J_N v = \lambda v, \qquad Bv = \zeta v, \qquad \beta = N\lambda/2.$$

Put $\alpha = (N^2 - 1)/4$. On $(-\pi/2, \pi/2)$ consider the Friedrichs realization

$$A_N u = -(\cos x\, u')' - \alpha \cos x\, u.$$

Its eigenfunctions are regular, with no logarithmic term, at both endpoints. Write its increasing eigenvalues as $\mu_{N,k}$, starting at $k = 0$, and define

$$Z_N = \sum_{k=N}^{\infty} \mu_{N,k}^{-1}. \tag{1}$$

The exact spectral-tail construction [1] gives

$$\{\mu_{N,0}, \dots, \mu_{N,N-1}\} = -\operatorname{spec} H, \qquad \mu_{N,N} > \|H\|, \tag{2}$$

and the norm-convergent scalar representation at every common mode

$$\log \frac{\zeta}{1-\zeta} = F_N(\beta), \qquad F_N(b) = 2 \sum_{k=N}^{\infty} \operatorname{artanh} \frac{b}{\mu_{N,k}}, \quad |b| < \mu_{N,N}. \tag{3}$$

For clarity, the ingredients of these exact inputs are as follows. The centered finite Fourier space is invariant under $-A_N$ and supplies $-\operatorname{spec} H$. A sign-change test against that space, together with the variational nodal bound, identifies exactly the first N differential eigenvalues. For the left regular solution normalized to one, let $C_N(b)$ be its logarithmic coefficient at the right endpoint.

Its Green kernel gives $C'_N/C_N = \mathrm{Tr}(A_N + b)^{-1}$. Deflating $D_N(b) = \det(bI - H)$ removes precisely those first N terms. The endpoint Lagrange identity and the complementary Fourier arc give the occupation ratio as $[C_N/D_N](\beta)/[C_N/D_N](-\beta)$. Integrating the deflated resolvent trace proves (3). These statements concern the specified endpoint realization, not an arbitrary self-adjoint extension.

At $b = 0$ set $\nu = (N-1)/2$ and translate $s = x + \pi/2$. The regular solutions are $P_\nu(\cos s)$ and $P_\nu(-\cos s)$, where P_ν denotes the Ferrers function on $(-1, 1)$. The Legendre connection formula gives $C_N(0) = 2\sin(\pi\nu)/\pi$, while the parameter Lagrange identity gives $C'_N(0) = -\int_0^\pi P_\nu(\cos s)P_\nu(-\cos s)\mathrm{d}s$. Because D_N is even and nonzero at zero,

$$Z_N = \frac{C'_N(0)}{C_N(0)} = -\frac{\pi}{2\sin\pi\nu}\int_0^\pi P_\nu(\cos s)P_\nu(-\cos s)\mathrm{d}s. \tag{4}$$

The notation $P_{-1/2}(\cosh t)$ below instead denotes the real Legendre function for argument at least one. All spectral and modular quantities here are dimensionless; no physical clock or energy scale is introduced.

2 A weighted resolvent and a Mellin reflection identity

Define

$$\Phi(z) = {}_2F_1(1/2, 1/2; 1; z), \qquad G(z) = \Phi(z)^2, \qquad c_j = \frac{(1/2)_j}{j!}, \qquad a_l = \sum_{j=0}^{l} c_j^2 c_{l-j}^2.$$

Thus $G(z) = \sum_{l\geq 0} a_l z^l$ for $|z| < 1$.

Lemma 1 (Weighted Legendre trace)**.** *Let*

$$M(a) = \sum_{l=0}^{\infty} \frac{a_l}{l + a}.$$

This defines a meromorphic function with possible poles only at the nonpositive integers. For even N, with $\nu = (N-1)/2$,

$$Z_N = \frac{\pi}{2}\{M(-\nu) + M(\nu + 1)\}. \tag{5}$$

For $\Re a > 0$, $M(a) = \int_0^1 z^{a-1}G(z)\mathrm{d}z$.

Proof. The generating function for Legendre polynomials yields

$$P_l(\cos s) = \sum_{j=0}^{l} c_j c_{l-j} e^{i(2j-l)s}, \qquad \int_0^\pi P_l(\cos s)^2 \mathrm{d}s = \pi a_l.$$

Induction gives $c_j^2 \leq 1/(j+1)$, and hence

$$0 < a_l \leq \frac{2H_{l+1}}{l+2}, \qquad H_r = \sum_{j=1}^{r} \frac{1}{j}. \tag{6}$$

Consider the ordinary Legendre operator with regular/Friedrichs endpoint conditions, $L = -[(1 - x^2)u']'$ on $L^2(-1, 1; \mathrm{d}x)$, with eigenvalues $l(l+1)$ and normalized eigenfunctions $\sqrt{(2l+1)/2}P_l(x)$. Its shifted Green kernel has diagonal

$$(L - \nu(\nu+1))^{-1}(x, x) = -\frac{\pi}{2\sin\pi\nu}P_\nu(x)P_\nu(-x).$$

Initially $-1/2 < \nu < 0$, so the resolvent is positive. Integrating its diagonal with weight $(1 - x^2)^{-1/2}$ gives, by positive-kernel trace on compact intervals and monotone exhaustion,

$$-\frac{\pi}{2\sin\pi\nu}\int_0^\pi P_\nu(\cos s)P_\nu(-\cos s)\mathrm{d}s = \frac{\pi}{2}\sum_{l\geq 0} a_l\left(\frac{1}{l-\nu} + \frac{1}{l+\nu+1}\right).$$

The series is locally normally convergent away from its poles by (6). Endpoint logarithmic bounds, locally uniform in ν, give the meromorphic continuation of the integral expression. Continuation to the required half-integer ν and (4) prove (5). The Mellin integral for M follows by absolute convergence. At negative half-integers it always means meromorphic continuation, not a divergent ordinary integral. □

Lemma 2 (Mellin reflection at half-integers)**.** *For $n \geq 1$,*

$$M(1/2-n) + M(1/2+n) = \int_0^1 z^{n-1/2} G(1-z)\mathrm{d}z. \tag{7}$$

Proof. For $0 < a < 1$, introduce

$$J(c) = \int_0^1 z^{c-1} G(1-z)\mathrm{d}z, \qquad Q(a) = \int_0^\infty x^{a-1} G(-x)\mathrm{d}x.$$

Use the principal branch of Φ, cut along $[1,\infty)$. Reciprocal-modulus continuation [3] gives, for $x > 1$,

$$\Phi(x+i0) = x^{-1/2}\{\Phi(1/x) + i\Phi(1-1/x)\}, \quad \Re G(x+i0) = x^{-1}\{G(1/x) - G(1-1/x)\}.$$

Integrate $z^{a-1}G(z)$ around the upper half-plane, with small indentations at zero and one. The origin arc is $O(\epsilon^a)$; the arc at one is $O(\epsilon(1+|\log\epsilon|)^2)$. Since $\Phi(z) = O(|z|^{-1/2}(1+\log|z|))$ there, the large arc is $O(R^{a-1}(1+\log R)^2)$ and also vanishes. The negative-axis phase therefore gives

$$\int_0^\infty x^{a-1} G(x+i0)\mathrm{d}x = e^{i\pi a} Q(a).$$

Taking real parts and changing $z = 1/x$ on $x > 1$ yields

$$M(a) + M(1-a) - J(1-a) = \cos(\pi a) Q(a). \tag{8}$$

Pfaff's transformation gives $G(-x) = (1+x)^{-1} G(x/(1+x))$, so

$$Q(a) = \int_0^1 z^{a-1}(1-z)^{-a} G(z)\mathrm{d}z.$$

For $\Re a < 1$, its endpoint at one is integrable. At zero, subtracting finitely many Taylor terms of $(1-z)^{-a}G(z)$ continues this integral meromorphically; its possible poles in that half-plane are only $0, -1, -2, \ldots$. Both sides of (8) consequently continue to $\Re a < 1$. At $a = 1/2 - n$, $Q(a)$ is finite and $\cos(\pi a) = 0$. This proves (7), including its normalization. □

3 Exact positive representation and trace asymptotics

Theorem 3 (Positive Laplace formula)**.** *For every positive even integer N,*

$$\boxed{Z_N = \pi \int_0^\infty e^{-Nt} f(t)\mathrm{d}t, \qquad f(t) = P_{-1/2}(\cosh t)^2.} \tag{9}$$

Equivalently, $Z_N = (\pi/2)\int_0^1 z^{(N-1)/2} G(1-z)\mathrm{d}z$.

Proof. The equivalent form follows immediately from (5) and (7). The quadratic transformation

$$y(t) := P_{-1/2}(\cosh t) = e^{-t/2}\Phi(1-e^{-2t})$$

then gives (9) on substituting $z = e^{-2t}$. The transformation can be checked directly: both sides are the unique regular solution of

$$y'' + \coth t\, y' + \tfrac{1}{4} y = 0, \qquad y(0) = 1, \quad y'(0) = 0. \tag{10}$$

Thus no numerical match or asymptotic normalization is required. □

Euler's integral gives the particularly useful real formula

$$y(t) = \frac{2}{\pi}\int_0^{\pi/2} \frac{\mathrm{d}\varphi}{\sqrt{1+\sinh^2(t/2)\sin^2\varphi}}. \tag{11}$$

It follows that $0 < f(t) \le 1$, with strict decrease for $t > 0$. The right side of (9) defines a completely monotone interpolation for real $N > 0$, but its identification with the spectral trace is asserted here only at even integers. In particular, odd N has a zero finite mode and requires a different deflation at $b = 0$.

Corollary 4 (An exact slope bound)**.** *For even $N \ge 2$,*

$$0 < \pi - NZ_N < \frac{\pi}{4N^2}. \tag{12}$$

Moreover NZ_N/π increases strictly along the positive even integers and tends to one.

Proof. Equation (10) implies

$$-y'(t) = \frac{1}{4\sinh t}\int_0^t \sinh s\, y(s)\mathrm{d}s < \tfrac{1}{4}\tanh(t/2) < t/8 \quad (t > 0).$$

Thus $0 < 1-f(t) = (1-y(t))(1+y(t)) < t^2/8$. Integration proves (12). Finally $NZ_N/\pi = \int_0^\infty e^{-u} f(u/N)\mathrm{d}u$; strict monotonicity and dominated convergence give the remaining claims. □

Theorem 5 (Asymptotic expansion with a remainder)**.** *As positive even $N \to \infty$,*

$$\boxed{Z_N = \frac{\pi}{N} - \frac{\pi}{4N^3} + \frac{17\pi}{64N^5} - \frac{169\pi}{256N^7} + \frac{50777\pi}{16384N^9} + O(N^{-11}).} \tag{13}$$

There is an ordinary asymptotic expansion to every fixed order. In (13), the absolute remainder is at most $\pi B\, 10!/N^{11}$, where

$$\rho = 3/2, \qquad M = (1-\sinh^2(3/4))^{-1},$$
$$B = \max\left\{\frac{M\rho^{-10}}{1-\rho^{-2}},\ 2 + \frac{1}{8} + \frac{17}{1536} + \frac{169}{184320} + \frac{50777}{660602880}\right\}.$$

Proof. Expansion of (10), or of its defining hypergeometric series, gives

$$f(t) = 1 - \frac{t^2}{8} + \frac{17t^4}{1536} - \frac{169t^6}{184320} + \frac{50777t^8}{660602880} + O(t^{10}). \tag{14}$$

For complex $|t| \le \rho$, $|\sinh(t/2)| \le \sinh(3/4) < 1$. Equation (11), with the square root continued from zero, shows that f is even and analytic on a neighborhood of this disk, with $|f| \le M$. Cauchy's estimate therefore bounds its coefficient of t^{2j} by $M\rho^{-2j}$. Let S_8 be the polynomial displayed in (14). For $0 \le t \le 1$, the geometric coefficient tail gives $|f(t) - S_8(t)| \le M\rho^{-10}t^{10}/(1-\rho^{-2})$. For $t \ge 1$, $0 < f \le 1$ gives the other term defining B. Thus $|f - S_8| \le Bt^{10}$ on the entire positive axis. Integrating this finite Taylor estimate in (9) proves the stated remainder and coefficients. The same argument applies at each fixed truncation order. It does not assert convergence of the infinite inverse-power series. □

The coefficient of N^{-9} is now a spectral-trace theorem. Its agreement with a formal quantum-period coefficient does not, by itself, prove that period expansion at other spectral modes.

4 A uniform estimate near the center

Define the leading reference function and errors by

$$\eta(t) = \pi t\, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2), \qquad \varepsilon = \log\frac{1-\zeta}{\zeta}, \qquad e = \varepsilon + N\eta(\lambda/N). \tag{15}$$

Theorem 6 (Signed plunge law for shrinking Jacobi modes)**.** *Let even $N \to \infty$ and choose nonzero common eigenmodes with $\lambda_N \to 0$. Then*

$$e_N = \frac{\pi\lambda_N}{4N^2}\left[1 + O(N^{-2} + \lambda_N^2)\right], \qquad \varepsilon_N = -\pi\lambda_N\left[1 + O(N^{-2})\right]. \tag{16}$$

Consequently

$$\boxed{\frac{-4N^2 e_N}{\varepsilon_N} \longrightarrow 1.} \tag{17}$$

The estimates apply to either sign of the eigenvalue.

Proof. We give a bound uniform for even $N \geq 8$ and $0 < |\lambda| \leq 1$. The central two-site Rayleigh quotient gives $\|H\| \geq N^2/8$. By (2),

$$r := \frac{|\beta|}{\mu_{N,N}} \leq \frac{4|\lambda|}{N} \leq \frac{1}{2}, \qquad \sum_{k\geq N} \mu_{N,k}^{-3} \leq \frac{64 Z_N}{N^4} \leq \frac{64\pi}{N^5}.$$

The positive artanh series in (3) gives

$$|F_N(\beta) - N\lambda Z_N| \leq \frac{2|\beta|^3}{3(1-r^2)} \sum_{k\geq N} \mu_{N,k}^{-3} \leq \frac{64\pi|\lambda|^3}{9N^2}. \tag{18}$$

The coefficients d_k of the hypergeometric series in (15) are positive, with $d_1 = 1/16$ and $d_{k+1} < d_k$. Hence, for $|t| < 1/2$,

$$|\eta(t) - \pi t| \leq \frac{\pi|t|^3}{4(1-4t^2)}, \qquad |N\eta(\lambda/N) - \pi\lambda| \leq \frac{4\pi|\lambda|^3}{15N^2}.$$

Since $\varepsilon = -F_N(\beta)$, these inequalities imply

$$|e - \lambda(\pi - NZ_N)| \leq \frac{8\pi|\lambda|^3}{N^2}. \tag{19}$$

Equation (13) gives $\pi - NZ_N = \pi/(4N^2) + O(N^{-4})$. This and (19) prove the first estimate in (16); (12) and (18) prove the second. Dividing is legitimate: (3) is strictly increasing and vanishes only at $\beta = 0$. No differentiation of an unproved spectral asymptotic has been used. □

5 The actual central pair satisfies the hypothesis

Lemma 7 (An elementary central-mode bound)**.** *Let $N = 2m$, $m \geq 1$, and let $\lambda_{\min,+}(J_{2m})$ be the smallest positive eigenvalue. With $H_0 = 0$,*

$$0 < \lambda_{\min,+}(J_{2m}) \leq \frac{2m-1}{2m}\left(1 + \frac{1}{4}H_{m-1}\right)^{-1/2}. \tag{20}$$

In particular the innermost positive and negative modes satisfy (16) *and* (17)*.*

Proof. Use site indices $1, \ldots, 2m$ and bonds $b_j = j(2m-j)/(2m)$. Set $v_{2k} = 0$, $v_{2k-1} = A_k$, $A_1 = 1$, and $A_{k+1} = -b_{2k-1}A_k/b_{2k}$. All rows of $J_{2m}v$ vanish except the last, so $\|J_{2m}v\| = b_{2m-1}|A_m|$. With $c_r = (1/2)_r/r!$ as above,

$$|A_k| = \prod_{j=1}^{k-1} \frac{2j-1}{2j}\frac{2m-2j+1}{2m-2j} \geq c_{k-1}, \qquad |A_m| = (2m-1)c_{m-1}^2 \leq 1.$$

For the last inequality, the ratio of successive endpoint amplitudes is $1-1/(4m^2) < 1$, starting at one. Also rc_r^2 increases for $r \geq 1$, starting at $1/4$, because its successive ratio is $(2r+1)^2/[4r(r+1)] > 1$. Thus $\|v\|^2 \geq 1 + H_{m-1}/4$. Expanding v in an orthonormal eigenbasis gives $\operatorname{dist}(0, \operatorname{spec} J_{2m}) \leq \|J_{2m}v\|/\|v\|$. Finally the spectrum is symmetric, and $\det J_{2m} = (-1)^m \prod_{k=1}^{m} b_{2k-1}^2 \neq 0$. The distance is therefore the smallest positive eigenvalue, proving (20). The harmonic sum diverges, so the required nonzero eigenvalues tend to zero. □

The estimate is sufficient but is not claimed to give the sharp central eigenvalue scale. In combination with the previous theorem it gives the conservative central-pair bound $-4N^2 e_N/\varepsilon_N = 1 + O(1/\log N)$.

6 Independent recurrence and scope

An additional exact characterization, useful for checking constants, is

$$(N+1)^3 Z_{N+2} - 2N(N^2+1)Z_N + (N-1)^3 Z_{N-2} = 0, \qquad N \geq 4 \text{ even.} \tag{21}$$

For a direct derivation from (4), let $I_\nu = \int_{-1}^{1} (1-x^2)^{-1/2} P_\nu(x) P_\nu(-x) \mathrm{d}x$ and insert an extra x^2 to define T_ν. The degree raising and lowering identities [3], followed by two integrations by parts, give

$$I_{\nu+1} = \frac{\nu}{(\nu+1)^2} T_\nu + \left[-1 + \frac{1}{2(\nu+1)^2}\right] I_\nu,$$
$$I_{\nu-1} = -\frac{\nu+1}{\nu^2} T_\nu + \left[-1 + \frac{1}{2\nu^2}\right] I_\nu.$$

Endpoint products are $O(r^{1/2}(1+|\log r|))$ and vanish. Eliminating T_ν gives

$$(\nu+1)^3 I_{\nu+1} + (2\nu+1)(\nu^2+\nu+1/2) I_\nu + \nu^3 I_{\nu-1} = 0.$$

Substitution in (4), including the alternating sine factor, proves (21). The recurrence alone would not have fixed the normalization of a candidate solution; the contour argument supplies that missing information.

Numerical evaluation of the connection derivative and both positive-integral forms agrees at $N = 2, 4, 8$ to 70-digit working precision. Exact rational checks verify the displayed kernel coefficients and trial-vector identities. These calculations support reproducibility; the proofs and remainder estimates above do not depend on finite numerical agreement.

The conclusions concern even size and half filling. They prove a reciprocal-trace expansion and the signed law for shrinking Jacobi modes, including the central pair. They do not establish the uniform bulk quantum-period remainder, its higher spectral coefficients, a general-filling effective-size convention, or the full Wilson relative-entropy finite-part asymptotic. The separate companion *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves the fixed-interior period expansion and leading Wilson law with bounded remainder.

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References

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[3] NIST Digital Library of Mathematical Functions, §14.3, §14.10, §15.6, §15.8, and §19.7; in particular (14.10.4)–(14.10.5), (15.6.1), (15.8.1), and (19.7.3).